EXPO
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TerraCrate

Secure, offline-first file storage for local networks

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Value Proposition

Existing file-sharing solutions force a trade-off between security and portability. USB drives lack access control, commercial NAS requires permanent infrastructure, and cloud platforms introduce third-party trust dependencies and internet requirements. No lightweight solution combines certificate based authentication, fine-grained access control, and zero configuration discovery in a single self contained device.

Objective

Build a portable, self-hosted file server on a Raspberry Pi that provides secure multi-user file sharing with mTLS authentication, three-tier permissions, and mDNS discovery, deployable in any environment without cloud infrastructure or client software beyond a browser.

Problem Definition

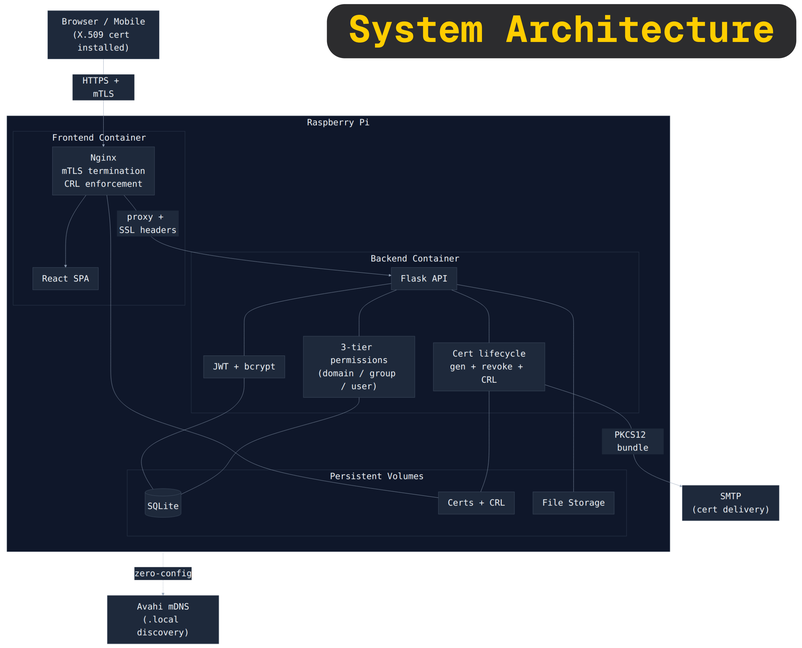
TerraCrate gives you full custody of your data in a device you can carry in your pocket. Plug it in and it is discoverable. Approve a user and they get a certificate in their inbox. Pull the plug and your data goes with you. No cloud, no residue, no trust required.

Key Requirements

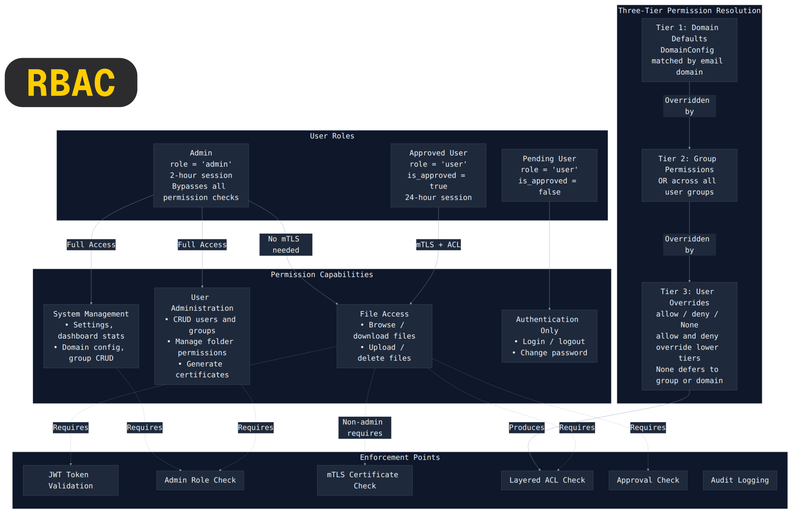
All communication must occur over HTTPS, with regular users authenticating via mutual TLS. The system must handle the full certificate lifecycle including generation, delivery via SMTP, and CRL-enforced revocation. A three-tier permission model spanning domain, group, and per-user overrides govern folder access. The device must advertise itself via mDNS with a Wi-Fi AP fallback for infrastructure less environments. All security relevant actions must be recorded in an audit log, and the entire system must run self-contained with no external dependencies.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION

System Architecture



RBAC



Concept Development

Two prior designs were evaluated: an SMB-over-mTLS approach requiring client-side Docker, and a QR code JWT approach lacking mutual TLS and per-user access control. The final design resolves both by delegating mTLS verification to Nginx, requiring only a browser with an installed certificate.

Validation

TerraCrate is validated through automated testing, static analysis, and security evaluation. The backend has 105 unit tests covering authentication, permission resolution, certificate lifecycle, and path traversal prevention, with security-critical invariants explicitly verified: token expiry, deny override logic, and CN mismatch triggered revocation. The frontend enforces quality through ESLint, TypeScript strict mode, and Vite build verification. A GitHub Actions pipeline enforces all checks on every push before code reaches the main branch. Security was evaluated against threats inherent to an untrusted wireless environment. mTLS provides replay and session hijacking resistance, bcrypt with short-lived JWTs limits credential exposure, and the CRL ensures immediate access termination on revocation. All administrative actions are captured in a queryable audit log supporting forensics and access accountability.

Conclusion

TerraCrate delivers a secure, portable, self-hosted file sharing system that operates entirely without cloud infrastructure. By delegating mTLS verification to Nginx, the system achieves cryptographic mutual authentication with no client-side software beyond a standard browser. The three-tier permission model, automated certificate lifecycle management, mDNS discovery, and comprehensive audit logging together satisfy all core requirements defined by the client. Future work includes clustered deployment, SMB filesystem access, peer-to-peer synchronization, attribute-based encryption, and anomaly detection on access logs.

Acknowledgements

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